

# Derivation of Wave Function Forms for Identical Particles from Particle Identity Uncertainty

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## Abstract

This paper derives the symmetrized wave function forms for identical particles starting from a classical picture of “particles with potentially distinguishable identities.” By rigorously applying the fundamental principle that “physical observations must be independent of particle identity,” we derive the necessary symmetrized forms of the wave function. This derivation clearly demonstrates the quantum origin of Bose and Fermi statistics, revealing the profound implication of the indistinguishability principle: it is not a limitation of experimental precision but an intrinsic requirement of physical laws.

## 1 Basic Assumptions and Mathematical Framework

### 1.1 Physical System

- There exist two **identical particles**.
- One is in the single-particle state  $\psi_a$ , the other in  $\psi_b$ .
- The particles are intrinsically **indistinguishable**.

### 1.2 Artificial Labels (Mathematical Tool)

For theoretical description, we artificially name the two particles  $\alpha$  and  $\beta$ .

**Crucial:** These labels have no physical meaning, are unmeasurable, and serve only as mathematical tools in the derivation.

### 1.3 Configuration Space Definitions

- **Single-particle position space:**  $R$  (elements:  $r_1, r_2, \dots$ )
- **Ordered configuration space for distinguishable particles:**  $R_\alpha \times R_\beta$  and  $R_\beta \times R_\alpha$   
(elements are ordered pairs  $(r_1, r_2)$ )
- **Physical configuration space (Symmetric product):**  $RS = (R \times R)/S_2$   
(elements are unordered pairs  $r_{12} = \{r_1, r_2\}$ )

## 1.4 Wave Function Definitions

- $\psi$ : Auxiliary wave function defined on the ordered configuration space.
- $\psi'$ : Physical wave function defined on the physical space  $RS$ .

Due to position indistinguishability, the physical wave function is defined as a coherent superposition:

$$\psi'(r_{12}) = \psi((r_1, r_2)) + \psi((r_2, r_1)) \quad (1)$$

## 2 First Hypothetical Case: $(\alpha@ \psi_a, \beta@ \psi_b)$

Assume particle  $\alpha$  is in state  $\psi_a$ , particle  $\beta$  is in state  $\psi_b$ .

Under this assumption, we compute the physical wave function  $\psi'_{(1)}(r_{12})$ . There are two indistinguishable paths leading to the measurement outcome  $r_{12}$ :

### 2.1 Path Analysis

- (i) Path  $\alpha@r_1, \beta@r_2$ 
  - Configuration:  $(r_1, r_2) \in R_\alpha \times R_\beta$
  - Probability amplitude density:  $\psi((r_1, r_2)) = A\psi_a(r_1)\psi_b(r_2)$
- (ii) Path  $\alpha@r_2, \beta@r_1$ 
  - Configuration:  $(r_1, r_2) \in R_\beta \times R_\alpha$  (denotes  $\beta@r_1, \alpha@r_2$ )
  - Probability amplitude density:  $\psi((r_2, r_1)) = B\psi_a(r_2)\psi_b(r_1)$

### 2.2 Coherent Superposition

According to quantum mechanics, probability amplitudes of indistinguishable paths must be **coherently superposed**:

$$\psi'_{(1)}(r_{12}) = A\psi_a(r_1)\psi_b(r_2) + B\psi_a(r_2)\psi_b(r_1) \quad (2)$$

## 3 Second Hypothetical Case: $(\alpha@ \psi_b, \beta@ \psi_a)$

Assume particle  $\alpha$  is in state  $\psi_b$ , particle  $\beta$  is in state  $\psi_a$ .

### 3.1 Path Analysis

- (i) Path  $\alpha@r_1, \beta@r_2$ 
  - Probability amplitude density:  $\psi((r_1, r_2)) = C\psi_b(r_1)\psi_a(r_2)$
- (ii) Path  $\alpha@r_2, \beta@r_1$ 
  - Probability amplitude density:  $\psi((r_2, r_1)) = D\psi_b(r_2)\psi_a(r_1)$

### 3.2 Coherent Superposition

The physical wave function for case (2) is:

$$\psi'_{(2)}(r_{12}) = C\psi_b(r_1)\psi_a(r_2) + D\psi_b(r_2)\psi_a(r_1) \quad (3)$$

## 4 Core Physical Constraint and Derivation

### 4.1 Identity Uncertainty Constraint

We now face **identity uncertainty**: Which is the actual case of the universe, (1) or (2)? We do not know.

**Key Constraint**: Since the particles are identical, any physical observation (probability density) must be independent of this identity assignment. Therefore:

$$|\psi'_{(1)}(r_{12})|^2 = |\psi'_{(2)}(r_{12})|^2 \quad \forall r_1, r_2 \quad (4)$$

**Important**: We do **not** take a linear combination of  $\psi'_{(1)}$  and  $\psi'_{(2)}$ , because the system is in one of them (classical mixture), not in a quantum superposition.

### 4.2 Algebraic Solution

To simplify calculations, set the phases of  $\psi_a$  and  $\psi_b$  to 1.

Substituting  $\psi'_{(1)}$  and  $\psi'_{(2)}$  into the constraint equation and comparing coefficients and phases yields:

#### 4.2.1 Coefficient Relations

$$|A| = |B|, \quad |C| = |D|, \quad |A| = |C| \quad (5)$$

$$\arg(A) - \arg(B) = \arg(D) - \arg(C) + 2n\pi \quad (6)$$

#### 4.2.2 Parametric Representation

Let:

$$A = ae^{i\theta_a}, \quad B = \pm ae^{i\theta_b} \quad (7)$$

### 4.3 Final Form

Substituting the coefficients back into  $\psi'_{(1)}$  (or  $\psi'_{(2)}$ , they are now equivalent) gives the general form of the total probability amplitude density:

$$\psi'_{\text{total}}(r_{12}) = ae^{i\theta_a}\psi_a(r_1)\psi_b(r_2) \pm ae^{i\theta_b}\psi_a(r_2)\psi_b(r_1) \quad (8)$$

### 4.4 Phase Simplification

- The global phase  $e^{i\theta_a}$  can be factored out and ignored.
- The relative phase  $e^{i(\theta_b - \theta_a)}$ , when taking the modulus squared:
  - If +1: No effect

- If  $-1$ : Produces a sign difference

Thus, the only physically significant forms are:

$$\psi'_{\text{total}}(r_{12}) = \mathcal{A} [\psi_a(r_1)\psi_b(r_2) \pm \psi_a(r_2)\psi_b(r_1)] \quad (9)$$

where  $\mathcal{A}$  is the normalization constant.

## 5 Conclusion

Starting from the classical picture of “particles with potentially distinguishable identities” and rigorously applying the constraint that “physical observations must be independent of identity,” we have derived:

### 5.1 Final Result

The **only physically allowed wave functions** describing two identical particles (in states  $\psi_a$  and  $\psi_b$  respectively) are:

**Bosons (symmetric form):**

$$\boxed{\psi'_{\text{Bose}}(r_{12}) = \mathcal{A} (\psi_a(r_1)\psi_b(r_2) + \psi_a(r_2)\psi_b(r_1))} \quad (10)$$

**Fermions (antisymmetric form):**

$$\boxed{\psi'_{\text{Fermi}}(r_{12}) = \mathcal{A} (\psi_a(r_1)\psi_b(r_2) - \psi_a(r_2)\psi_b(r_1))} \quad (11)$$

### 5.2 Physical Significance

This derivation demonstrates that:

1. The indistinguishability of identical particles is not a limitation of experimental precision but an intrinsic requirement of physical laws.
2. To ensure the determinism of physical measurements, the quantum state must possess a specific symmetric structure.
3. It naturally leads to the statistical rules for bosons and fermions.
4. It shows the necessary transition from classical concepts to quantum description.